

Leveraging Machine Learning Techniques in Enhancing Recognition of Emotion in Speech

Ghanisht Aggarwal*, Yash Khanna

ghanisht.aggarwal@gmail.com*, yashkhanna@ieee.org

Maharaja Agrasen Institution of Technology, Rohini, New Delhi, India, 110086

Abstract: Emotions can play an important role in determining how we think and behave. Emotions compel us to take decisions be it small or big. In order to understand emotions, it is paramount that we understand the critical expressive component. While interacting with people, it is cardinal to provide clues in the form of emotions to interpret and react accordingly. In this work, to tackle the ambiguity of speech, we have adopted an engineering technique based on speech emotion recognition. Formalizing our concern as a multi-class classification model, we compare the performances of different machine learning models by extracting numerable artisanal features of the audio signal and employing them to train six conventional machine learning models. For the various experiment settings in which we tested our models, we document accuracy, f-score, accuracy and recall. We are able to achieve at par performances form Gradient boosting and Random Forest classifiers. Ultimately, we have shown that simpler machine learning based models trained over a few hand-crafted features are able to achieve performances that may be analogous to the current deep learning based state-of-the-art methods.

Lakshminarasimhan, P. P. Liang, A. Zadeh and L.P. Morency, 2018) are among the recent proposals to help take SER forward.

1. Introduction

Correspondence is indispensable for human presence and as a rule, we need to manage questionable circumstances. For instance, the expression "This is great" could be said under either upbeat or miserable settings. People can resolve uncertainty by and large since we can productively grasp data from various spheres such as discourse, audio and visual. With the ascent of profound learning calculations, there have been different endeavours to handle the assignment of Speech Emotion Recognition (SER) as in (Z. Huang, M. Dong, Q. Mao and Y. Zhan, 2014) (S. Yoon, S. Byun and K. Jung, 2018) (K. Han, D. Yu and I. Tashev, 2014). Nonetheless, this ascent has made specialists spend more on the intensity of the profound learning models rather than utilizing area information to develop important highlights and build better-performing and better-interpretable models. In this work, we investigate the ramifications of hand-made highlights for SER and analyse the presentation of lighter Artificial Intelligence (AI) models with the vigorously information dependent profound learning models (G. Sahu and D. R. Cheriton). Moreover, we additionally join highlights from the printed methodology to comprehend the relationship between various modalities and help vagueness goals. All the more officially, we represent our errand as a multi-class arrangement issue and utilize the two classes of models to explain that. For both the methodologies, we first concentrate hand-created highlights from the time area of the sound sign and train the separate models.

We train traditional Machine Learning (ML) classifiers namely Logistic Regression, Random Forests, Support Vector Machines, Gradient Boosting and Naive-Bayes. The models are evaluated on the Toronto Emotional Speech Set (TESS) dataset under different settings namely audio-only.

2. Literature Review

In this part, we study a portion of the efforts that have been made in the discipline of Speech Emotion Recognition (SER). The undertaking of SER isn't novel and has been read for a long period of time in literature. Utilizing Hidden Markov Models (HMMs) (L. R. Rabiner and B.H. Juang, 1986) are a dominant part of the early methodologies (A. Nogueiras, A. Moreno, A. Bonafonte and J. B. Mariño, 2001) (B. Schuller, G. Rigoll and M. Lang, 2003) for distinguishing feeling from discourse. State-of-the-art performances have been observed after the advent of deep neural networks. The amalgamation of features such as Tensor Fusion Networks (Z. Liu, Y. Shen, V. B. Lakshminarasimhan, P. P. Liang, A. Zadeh and L.P. Morency, 2017) and Low-Rank Matrix Multiplication (Z. Liu, Y. Shen, V. B.

This work aims to provide an analysis of lighter machine learning models trained over features that help in reducing obscurity in SER.

3. Dataset

In this work, we use the TESS (M. K. P. Kate Dupuis) released in 2010 by researchers at the University of Toronto. A set of 200 target words were spoken in the carrier phrase "Say the word _____" by two actresses (aged 26 and 64 years) and recordings were made of the set portraying each of seven emotions (anger, disgust, fear, happiness, pleasant surprise, sadness, and neutral). There are 2800 stimuli in total. Audiometric testing indicated that both actresses have thresholds within the normal range.

4. Methodology

4.1. Data Pre-processing

Audio: The above dataset for initial analysis offers an insight into the balancing of the dataset. In fact, there are two voices, a young female and an older female. All the information was preprocessed and sample-by-sample standardized, since there were not so many samples, and standardizing across samples would make it difficult to discern the data across classes with little variation. The normalization was based on loudness and length, for which we did zero padding to the heads and tails of all samples to make them length invariant.

4.2. Feature Extraction

The features used to train both models, the ML-based models, are now defined.

(a) *Pitch:* Wave-structures delivered by our vocal cords differ based on our emotion. Pitch is one of the features that help us recognize the same. Although there have been numerous calculations for assessing the pitch signal, we put our trust on utilizing the most widely recognized technique dependent on autocorrelation of centre-clipped frames (M. Sondhi, 1968). Formally, the input signal $y[n]$ is center-clipped to give a resultant signal, $y_{clipped}[n]$:

$$y_{clipped}[n] = \begin{cases} y[n] - C_l, & \text{if } y[n] \geq C_l \\ 0, & \text{if } |y[n]| < C_l \\ y[n] + C_l, & \text{if } y[n] \leq -C_l \end{cases} \quad (1)$$

C_l is typically almost half the average of the input signal. The distinct nature of input signal is denoted by $[\cdot]$. The autocorrelation is now calculated for the $y_{clipped}$ signal received, which is further normalised, and the peak values associated with the $y[n]$ input pitch given. Centre-clipping of the input signal was observed to result in more distinct peaks of autocorrelation.

(b) *Harmonics*: Other than pitch, there are extra excitation signs (H. Teager and S. Teager, 1990) (G. Zhou, J. H. Hansen, and J. F. Kaiser, 2001), in the emotional state of frustration or for stressed expression. The spectrum displays this additional excitation as harmonics (see Figure 1) and cross-harmonics. The harmonics calculation is carried out using a median-based filter as defined in (D. Fitzgerald, 2010). First, for a given window size l , the median filter is generated, given by:

$$y[n] = \text{median}(x[n-k : n+k] | k = (l-1)/2) \quad (2)$$

l is an odd value here. Considering the cases where l is supposed to be even, at the midpoint of the organised list, the midpoint is achieved as the mean of two values. Applying this filter to S_n , which is the h^{th} frequency slice of a given spectrogram S , to get a harmonic-enhanced spectrogram frequency slice H^h as:

$$H_i = M(S_h, l_{\text{harm}}) \quad (3)$$

Where the midpoint filter is M , i is the i^{th} time step and the length of the harmonic filter is l_{harm} .

(c) *Speech Energy*: Since a speech signal's energy can be linked to how loud it is, we utilize the same to study certain feelings. To reflect speech energy, we use the conventional Root Mean Square Energy (RMSE) which is given by the equation:

$$E = \sqrt{\frac{1}{n} \sum_{i=1}^n y[i]^2} \quad (4)$$

RMSE is calculated frame by frame and we take statistical features such as the mean and standard deviation.

(d) *Pause*: In order to reflect the "silent" part of the audio signal, we use this function. This amount is directly linked to our feelings; for example, when excited (say, angry or happy, resulting in a low pause value), we can appear to talk very quickly. Value for the function is given by:

$$\text{Pause} = \text{Pr}(y[n] < t) \quad (5)$$

where t is represented by a threshold of $\approx 0.4 * E$, E being the RMSE.

(e) *Central moments*: Ultimately, we use the mean and standard deviation of the signal amplitude to integrate the input's "summarised" information.

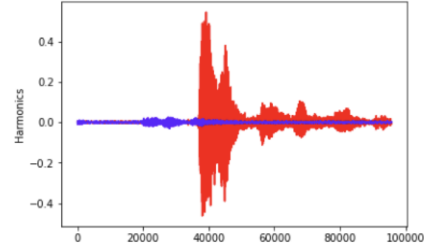


Fig. 1: Harmonics of angry (red) and sad (blue) audio signals

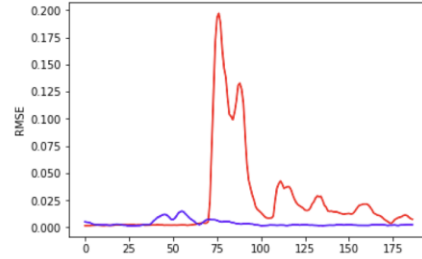


Fig. 2: RMSE plots of angry (red) and sad (blue) audio signals

4.3 Machine Learning Models

Following different ML-based classifiers are considered in this work.

1] *Random Forest (RF)*: Random forests are ensemble classifiers that curate multiple decision trees at the point of training and yield the class that is the classification of the individual trees' groups. It has two working standards that are basic:

- Each choice tree predicts utilizing an arbitrary subset of highlights (Y. Amit, D. Geman, and K. Wilder, 1997).
- With only a subset of preparation checks, each selection tree is prepared. This is regarded as conglomerating bootstraps (L. Breiman, 1996).

Finally, to predict the category of a given input, the majority vote of all the decision trees is taken.

2] *Gradient Boosting (XGB)*: eXtreme Gradient Boosting is a boosting technique that supports a quick preparation of the model in a parallelized way. Another ensemble classifier that combines a number of weak learners,

usually decision trees, is Boosting. Unlike RFs, they are educated in a successive manner, using forward-stagewise additive modelling. The decision trees learned during the early iterations are straightforward. When the classifier is made to focus on the instances highlighting the mistakes made by the previous learners, it marks a remarkable milestone. Once all the features are trained, we are ultimately provided with a prediction that is a weighted linear combination of the output from the individual learners (J. Friedman, T. Hastie, and R. Tibshirani, 2001).

3] *Support Vector Machines (SVMs)*: SVMs are in the bracket of supervised learning models consisting of correlated training algorithms. This helps in analysing data and solving problems related to data analysis, problems that are branched under regression and classification. When learning using the SVM training algorithm a non-probabilistic linear binary classifier is constructed (although methods such as Platt scaling (J. Platt, 1999) exist to use SVM in a probabilistic classification setting). It represents each preparation model as a point in space, planned with the end goal that the instances of the different classes are separated by a reasonable gap that is as wide as could be expected under the circumstances (which is achieved by minimizing the hinge loss). New models are then plotted into that equivalent space and anticipated to have a place with a classifier depending on which side of the gap they fall. SVMs were initially programmed to perform linear classification; however, they can effectively break down a nonlinear classification using the kernel trick (C. Cortes and V. Vapnik, 1995).

4] *Multinomial Naive Bayes (MNB)*: Naive Bayes classifiers are considered to be simple "probabilistic classifiers", based on the theorem of Bayes with clear assumptions of independence between the characteristics. The feature vectors reflect the frequencies at which various events have been produced by a multinomial ($p_1, \dots, p_n$) under multinomial settings, where p_i is the probability of an event occurring. Multinomial distribution defines the probability of observing counts among a number of groups, and thus multinomial naive Bayes is most suitable for features that reflect counts or count rates. MNB is very common for textual document classification tasks (A. M. Kibriya, E. Frank, B. Pfahringer, and G. Holmes, 2004) which is also essentially a problem of multi-class classification.

5] *Logistic Regression (LR)*: Problems with binary classification (G. King and L. Zeng, 2001) are solved using LR, that is, when we only have two labels. Logistic Regression is used when the dependent variable (target) is categorical. LR is implemented in this work in a one-vs-rest manner. We trained six classifiers for each class, and finally, took into consideration the class with the highest likelihood that was initially expected. We trained the above classifiers on state-of-the-art emotion recognition TESS dataset.

4.4 Experiments

For the audio environment, we train all classifiers using only the previously mentioned audio function vectors.

4.5 Implementation Details

We explain the implementation specifics utilised in this work in this section.

- In order to process the audio files and extract features from them, we use librosa (B. McFee, C. Raffel, D. Liang, D. P. Ellis,

M. McVicar, E. Battenberg, and O. Nieto, 2015) a Python library.

- Implementation of all the ML classifiers (XGB, SVM, LR, RF and MNB), were using scikit-learn and XGBoost (F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, 2011) (T. Chen, 2014) the machine learning libraries for Python.

We have split our dataset randomly into a train (80%) and a test (20%) package. For all the tests, the same split is used to guarantee a fair comparison. Cross-validation values were also considered to cross verify the same.

4.6 Evaluation Metrics

(a) *Accuracy*: This refers to the percentage of correctly graded research samples.

(b) *Precision*: Of all forecasts, this test informs us how many are actually present in the ground reality (a.k.a. labels).

(c) *Recall*: This test informs us how many right labels in the expected output are present.

Here, t_p , f_p , and f_n respectively stand for true positive, false positive, and false negative. From these values from the confusion matrix, we calculate these values.

(d) *F-score*: It is defined as the harmonic mean of precision and recall. This metric was used since precision is not a full measure of the predictive ability of a model, but because it is more normalised.

5. Results

Table 1

Models	Accuracy	F1	Recall	Precision
RF	56.8	56.0	57.3	57.2
XGB	56.7	56.0	56.9	57.0
MNB	32.0	11.0	17.0	20.4
LR	33.6	14.9	20.4	18.7
SVM	38.0	16.2	22.1	30.1

Fig. 3: Performance of different models

We can observe from the above Figure 3, that the light machine learning models have performed well and maybe comparable to the results from various heavy state-of-the-art deep neural networks. We reported the above metrics using optimal hyperparameter tuning.

6. Conclusion and Future Work

In this research endeavour, we have discussed the task of speech emotion recognition and further worked on the contribution of various modalities for the resolution of uncertainty on the TESS dataset. We study ML-models and illustrate that good performance can be achieved by even simpler and

clearer ML models. To resolve the formidable task of reducing ambiguity from the speech dataset, we used an engineering technique based on speech emotion recognition and curated our problem into a multi-class classification problem by extracting hand-crafted time-domain audio features. We can further exhibit that combining multiple ML models or using state-of-the-art LSTMs (S. Hoch Reiter and J. Schmid Huber, 1997) also contribute to some enhancement in the performance. We can create a comparison between both the ML- and DL- based techniques and identify if light weighted machine learning models are at par with the heavy deep neural networks. We just derive from audio signals a handful of time-domain characteristics. In order to make the audio features even more rich and functional, we could also implement various frequency-domain features such as Mel-Frequency Cepstral Coefficients (MFCC) (S. Davis and P. Mermelstein, 1980) Spectral Roll-off and additional time-domain features such as Zero Crossing Rate (ZCR) (F. Gouyon, F. Pachet, O. Delerue, 2000). Better fusion approaches such as TFN (A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.P. Morency, 2017) and LMF (Z. Liu, Y. Shen, V. B. Lakshmi Narasimhan, P. P. Liang, A. Zadeh, and L.P. Morency, 2018) may also be used to more reliably combine text and speech vectors. Including more data may as well help scale the performance of ML models.

7. References

- [1] Z. Huang, M. Dong, Q. Mao, and Y. Zhan, "Speech emotion recognition using CNN," in *Proceedings of the 22nd ACM international conference on Multimedia*, pp. 801–804, ACM, 2014.
- [2] S. Yoon, S. Byun, and K. Jung, "Multimodal speech emotion recognition using audio and text," in *2018 IEEE Spoken Language Technology Workshop (SLT)*, pp. 112–118, IEEE, 2018.
- [3] K. Han, D. Yu, and I. Tashev, "Speech emotion recognition using deep neural networks and extreme learning machines," in the *Fifteenth annual conference of the international speech communication association*, 2014.
- [4] S. Hoch Reiter and J. Schmid Huber, "Long short-term memory," *Neural computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [5] M. K. P. Kate Dupuis, "Toronto emotional speech set (TESS)," 2010. [Online]. Available: <https://tspace.library.utoronto.ca/handle/1807/24487>
- [6] A. Nogueiras, A. Moreno, A. Bonafonte, and J. B. Mariño, "Speech emotion recognition using hidden Markov models," in *Seventh European Conference on Speech Communication and Technology*, 2001.
- [7] B. Schuller, G. Rigoll, and M. Lang, "Hidden Markov model-based speech emotion recognition," in *2003 IEEE International Conference on Acoustics, Speech, and Signal Processing, 2003. Proceedings. (ICASSP 03)*, vol. 2, pp. II– 1, IEEE, 2003.
- [8] L. R. Rabiner and B.H. Juang, "An introduction to hidden Markov models," *IEEE ASSP magazine*, vol. 3, no. 1, pp. 4– 16, 1986.
- [9] Y. Kim, H. Lee, and E. M. Provost, "Deep learning for robust feature generation in audiovisual emotion recognition," in *2013 IEEE International Conference on Acoustics, Speech and Signal Processing*, pp. 3687–3691, IEEE, 2013.
- [10] A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.P. Morency, "Tensor fusion network for multimodal sentiment analysis," *arXiv preprint arXiv:1707.07250*, 2017.
- [11] Z. Liu, Y. Shen, V. B. Lakshmi Narasimhan, P. P. Liang, A. Zadeh, and L.P. Morency, "Efficient low-rank multi-modal fusion with modality-specific factors," *arXiv preprint arXiv:1806.00064*, 2018.
- [12] M. Sondhi, "New methods of pitch extraction," *IEEE Transactions on audio and electroacoustics*, vol. 16, no. 2, pp. 262–266, 1968.
- [13] H. Teager and S. Teager, "Evidence for nonlinear sound production mechanisms in the vocal tract," in *Speech production and speech modelling*, pp. 241–261, Springer, 1990.
- [14] G. Zhou, J. H. Hansen, and J. F. Kaiser, "Nonlinear feature-based classification of speech under stress," *IEEE Transactions on speech and audio processing*, vol. 9, no. 3, pp. 201– 216, 2001.
- [15] D. Fitzgerald, "Harmonic/percussive separation using median filtering," 2010.
- [16] Y. Amit, D. Geman, and K. Wilder, "Joint induction of shape features and tree classifiers," *IEEE Transactions on Pattern Analysis & Machine Intelligence*, no. 11, pp. 1300– 1305, 1997.
- [17] L. Breiman, "Bagging predictors," *Machine learning*, vol. 24, no. 2, pp. 123–140, 1996.
- [18] J. Friedman, T. Hastie, and R. Tibshirani, *The elements of statistical learning*, vol. 1. Springer series in statistics New York, 2001.
- [19] J. Platt *et al.*, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," *Advances in large margin classifiers*, vol. 10, no. 3, pp. 61– 74, 1999.
- [20] C. Cortes and V. Vapnik, "Support-vector networks," *Machine learning*, vol. 20, no. 3, pp. 273–297, 1995.
- [21] A. M. Kibriya, E. Frank, B. Pfahringer, and G. Holmes, "Multinomial naive bayes for text categorization revisited," in *Australasian Joint Conference on Artificial Intelligence*, pp. 488–499, Springer, 2004.
- [22] G. King and L. Zeng, "Logistic regression in rare events data," *Political analysis*, vol. 9, no. 2, pp. 137–163, 2001. [23] B. McFee, C. Raffel, D. Liang, D. P. Ellis, M. McVicar,
- [23] E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in python," in *Proceedings of the 14th python in science conference*, pp. 18–25, 2015.
- [24] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, *et al.*, "Scikit-learn: Machine learning in python," No. Oct, pp. 2825–2830, 2011.

[25] T. Chen, “Scalable, portable and distributed gradient boosting (gbdt, gbdt or gbm) library for Python, R, Java, Scala, C++ and more. Runs on single machine, Hadoop, Spark, Flink and Dataflow,” 2014.

[26] S. Davis and P. Mermelstein, “Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences,” *IEEE transactions on acoustics, speech, and signal processing*, vol. 28, no. 4, pp. 357–366, 1980.

[27] F. Gouyon, F. Pachet, O. Delerue, *et al.*, “On the use of zero-crossing rate for an application of classification of percussive sounds,” in *Proceedings of the COST G-6 conference on Digital Audio Effects (DAFX-00)*, Verona, Italy, 2000.

[28] G. Sahu and D. R. Cheriton, “Multimodal Speech Emotion Recognition and Ambiguity Resolution,” *Tech. Rep. [Online]. Available: <http://tinyurl.com/y55dlc3m>*